01 / PROBLEM AND AUDIENCE

Rent Radar

A rental map for Harvard and MIT students that shows what a listing leaves out.

A listing shows rent and photos. It does not answer:

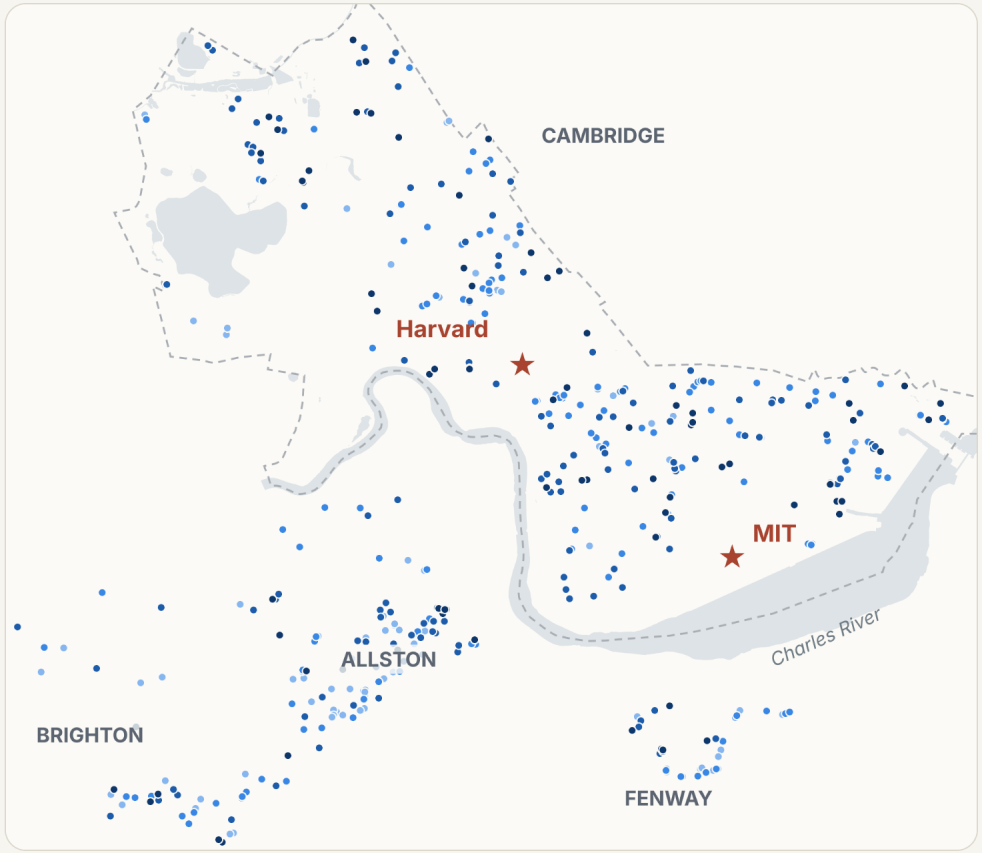
- Is the street safe to walk or bike at night?
- How long is the trip to campus, really?
- Do people report rats, noise, or no heat here?

For: students looking for an apartment in Allston, Brighton, Fenway, or Cambridge. Many are new to Boston, many come from abroad, and most rent sight unseen.

Hypothesis: one map with evidence for each listing helps students rule out bad apartments faster.

02 / PROPOSED SOLUTION

One map, one click, one report



Monthly rent, real listings

≤ $2,500 $2,501–3,500 $3,501–5,000 > $5,000

Browse

See apartments for rent right now. Filter by rent, bedrooms, and area.

Check

Click a listing: crashes, commute, 311 complaints, restaurants, each linked to its record.

Ask

“Under $2,500, near MIT, quiet.” The agent turns this into filters, then shows the top options and why.

Map: 731 RentCast listings in student areas, pulled Oct 27, 2026 (sample of 1,000)

03 / APPROACH

Public data in, evidence out

Data

- RentCast listings API
- Boston 311 requests
- MassDOT crashes
- Food inspections
- MBTA transit data

→

Pipeline

- Python fetch, weekly
- Validate and clean
- Join by location
- Store in SQLite

→

AI

- Turn 311 text into signals: no heat, pests, noise
- Agent turns plain words into filters and calls tools
- Answers cite records

→

Website

- Map of listings
- One-click report
- Ask box

Why not just ask a chatbot? Thousands of live listings and about 287,000 Boston 311 requests a year do not fit in a chat. The numbers must be computed and traceable.

Resources: RentCast API (free, 50 calls/mo) · OpenAI model with tool calling · Boston open data · Python, SQLite, Leaflet

311 volume: 287,148 requests in 2024 through Dec 10 (WBUR)

Start small: four student areas

In version 1

- Allston, Brighton, Fenway, Cambridge
- Rent, crashes, 311, commute, restaurants
- Map, report, and agent
- About 10 API calls per weekly refresh

Not in version 1

- The rest of Boston
- Photos and listing text (the API has none)
- Contacting landlords

Risk	Evidence	Plan
Stale listings	63% of sampled listings were listed 180+ days ago	Show days on market, flag old ones, spot-check
API limit	Free plan: 50 calls a month	Four areas only, weekly refresh, cache in SQLite
Two data systems	Cambridge data is not on Boston's portal	Boston layers first, then Cambridge
AI makes things up	An agent can state facts it never checked	Answer only from tool results; cite every number
Fair housing	Area scores can hide bias	No demographic data; show facts, not "good" or "bad" areas

Evidence: my RentCast test pull, Oct 27, 2026 (735 listings in the four areas)

05 / PLAN AND SUCCESS CRITERIA

Four steps, four measurable targets

BY THE MIDPOINT

1. **Data pipeline + map**
   Listings, 311, crashes in SQLite + map

2. **Listing report**
   Click-a-listing report; 311 classifier

BY THE FINAL

3. **Agent**
   Tools, citations, guardrails; 20 tests

4. **Polish and test**
   5 student testers, README, final demo

100%
of active listings in the four areas are on the map, refreshed weekly

≥ 85%
agreement between the 311 classifier and my labels on 200 complaints

0
made-up numbers in 20 agent test questions: every claim cites a record

5
student testers rule out bad listings faster than with listing sites alone

Targets are proposed; I will adjust them after feedback

APPENDIX / DATA CHECK

The listing data works: test pull, Oct 27

RentCast API, active rentals

Field	Boston	Cambridge
Active listings (API total)	7,292	1,193
Pulled for this test	500	500
Address, location, rent, beds	100%	100%
Square footage	79%	89%
Agent contact	16%	16%
Description or photos	none	none

Student areas in the sample

Area	Listings	Median	1-bed
Allston	94	$2,838	$2,325
Brighton	82	$3,123	$2,325
Fenway	59	$3,300	$2,700
Cambridge	500	$3,600	$3,209

Data quality: 5 Cambridge rows had a wrong ZIP code but a correct location, so the map places listings by coordinates.

Sample: first 500 active listings per city, sorted by last-seen date. Monthly asking rent.